Functions

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- Output:
  char ch = 'A'; → ASCII
  int x = ch;
  System.out.println(x); → 65
- Output:
  System.out.println('A'+1); → 66

- Divide 10 by 3 and roundoff the answer upto 3 decimal Places.
- Decimal to binary
- Write the Prime Factorization Of a number.

long a = 6;
int b = 4;
↳ Smaller datatype.

b = a; (x)

3/2.0
↳

(3/2) → Integer division

(floor) (3/2)? → Int
↳ 0.0

char a = 'a'
↳ int b = a; → ASCII

'a' → 97

b = a
b = 97

'A' = 65 , 'B' = 66 , 'C' = 67

→ 'A' + 1

'A' = 65
'B' = 66

→ Char + int

$\hookrightarrow 65 + 1 = 66$

Task:- Print :-

A to Z (Don't use numbers in the loop)

```
for (char i = 'a' ; i <= 'z' ; i++){
    Sout(i);
}
```

int

$10/3 = 3.333$

3.3333

```
float a = 10/3 ;
Math.round(a × 1000)/1000.0
```

3.33333

3333.33

3333.333

3.333

.5

3334

Math.round(3333.59) = 3334

<.05

3333.3

3333

3333/1000.0

3333/1000.0 → float / double

3.333

→ Binary to Decimal

11001 17

5 4 3 2 1 0

$\sum \text{Digit} \times 2^{\text{position}}$

$1\times 2^0 + 1\times 2^1 + 0\times 2^2$
$+ 0\times 2^3 + 1\times 2^4 + 1\times 2^5$
$= 51$

2	51	1
2	25	1
2	12	0
2	6	0
2	3	1
	1	

2)51(25
50
1

110011

2	95	1
2	47	1
2	23	1
2	11	1
2	5	1
2	2	0
	1	

2)95(47
8
15
14
1

$32+16+8+4$
$+2+1$

4 3 2 1 0
1 0 1 1 1 1 1

$64+16+8+4+2+1$
$= 95$

1 0 1 1 1 1 1
6 5 4 3 2 1 0

n = 23

Reverse

1 0 1 1 1

2	23	1
2	11	1
2	5	1
2	2	0
	1	

11101

1	0	1	1	1
4	3	2	1	0

$1\times 2^0 + 1\times 2^1 + 0\times 2^2 + 1\times 2^3 + 1\times 2^4$

$16 + 4 + 2 + 1$

$= 23$

0 →*

10 1 1 1

```
int n = 10;
int ans = 0; // binary number.

while (n>0){
    int rem = n%2;
    ans = ans*10 + rem;
    n = n/2;
}
```

→ 0, 1

11

23/2 = 1

ans = 0 + 1
ans = 1

n = 11

11/2 = 1

ans = 1×10 + 1 = 11

n = 5/2 = 1

11×10 + 1
= 111

2/2 = 0

ans = 111×10 + 0 = 1110

n = 1/2 = 1

1110×10+1
→ 11101

→ 10111 Ans.

18 → 2×3×3

Prime factorization

2	18
3	9
3	3
	1

16 = 2×2×2×2

21/1 = 0

21

1 2 3 4 5 6 7 8 9 - - - 21

21/2 21/3 = 7 7/7 = 1

$\cancel{7/3}$

$\sqrt{21}$

$= 4 : \ldots$

$= 4$

$1 - n$

$1 - \sqrt{n}$

2 3 4

3 × 7

$\cancel{21/2}$

$21/3$ ✓ $\boxed{n = 7}$

3 × 7 ✓ ✓

$n = n/3 = \cancel{7/3}$ ✗ ✓

$7/4$ ✗

18 $\sqrt{18} = 4$

2 3 4

$18/2 == 0$ → $\boxed{2 \quad 3 \quad 3}$

$10/2 \; \cancel{= 9/2}$ ✗

$\boxed{9/3 == 0}$ ✗

$3/3 == 0$

$n = 1 \% 3$ ✗

$1 \% 4 == 0$ ✗

16

$i = 2 \quad 3 \quad 4$

$16/2 = 8/2 = 4/2$

$== 0$ $= 2/2$

$i - \sqrt{n}$ Sq. $i^2 = n$

$i \times i = n$

→ if $(n \% i == 0)$

```
for( int i = 2 ; i*i <= n ; i++){
    while (n % i == 0) {
        Sout( i + " x ");
        n = n/i ;
    }
}
```

16 → 8

8 → 4 → 2 → 2 ✓

→ 2 × 2

× 2

× 2

$21/3 = 7$

3 ×

```
if(n%3==0)
    cout(n);
}
```

21 → 2 3 4

21 % 3 == 0

<u>function</u> → A Block of Code, with a return value that can be used again & again in a program.

How to define a function.

```
datatype function_name( ){
    - - - -
    - - - -
    - - - -
}
```

```
int add (int a, int b){
    int sum = a+b;
    return sum;
}

main(){
    add(2,3);
}
```